

Creating a Dataset

Aim:

To create a music dataset with missing values and handle them using mean and mode in Python.

Procedure:

1. Import required libraries (pandas, numpy).
2. Create a dataset with missing values using a dictionary.
3. Convert the dictionary to a Pandas DataFrame.
4. Display the original dataset.
5. Find missing values using `isnull().sum()`.
6. Replace missing numeric values with the mean using `fillna()`.
7. Replace missing categorical values with the mode.
8. Update the dataset using `inplace=True`.
9. Display the cleaned dataset.
10. Verify that all missing values are filled.

Program:

```
import pandas as pd
import numpy as np

data = {
'Song': ['Shape of You', 'Blinding Lights', 'Levitating', 'Peaches', 'Stay'],
'Artist': ['Ed Sheeran', 'The Weeknd', 'Dua Lipa', np.nan, 'Justin
Bieber'],
```

```

'Genre': ['Pop', 'Synthwave', np.nan, 'R&B', 'Pop'],
'Duration': [4.2, 3.5, np.nan, 3.1, 2.8],
'Streams': [2500, np.nan, 1800, 1200, 2700]
}

df = pd.DataFrame(data)

print("Original Dataset:\n", df)

print("\nMissing Values Count:\n", df.isnull().sum())

df['Duration'].fillna(df['Duration'].mean(), inplace=True)
df['Streams'].fillna(df['Streams'].mean(), inplace=True)
df['Artist'].fillna(df['Artist'].mode()[0], inplace=True)
df['Genre'].fillna(df['Genre'].mode()[0], inplace=True)

print("\nAfter Handling Missing Values:\n", df)

```

Output:

Original Dataset:

	Song	Artist	Genre	Duration	Streams
0	Shape of You	Ed Sheeran	Pop	4.2	2500.0
1	Blinding Lights	The Weeknd	Synthwave	3.5	NaN
2	Levitating	Dua Lipa	NaN	NaN	1800.0
3	Peaches	NaN	R&B	3.1	1200.0
4	Stay	Justin Bieber	Pop	2.8	2700.0

Missing Values Count:

Song	0
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Artist 1
Genre 1
Duration 1
Streams 1
dtype: int64

After Handling Missing Values:

	Song	Artist	Genre	Duration	Streams
0	Shape of You	Ed Sheeran	Pop	4.200000	2500.0
1	Blinding Lights	The Weeknd	Synthwave	3.500000	2050.0
2	Levitating	Dua Lipa	Pop	3.400000	1800.0
3	Peaches	Ed Sheeran	R&B	3.100000	1200.0
4	Stay	Justin Bieber	Pop	2.800000	2700.0

Result:

Hence, a Python program to create and preprocess a music dataset by filling missing values using mean and mode is successfully executed.